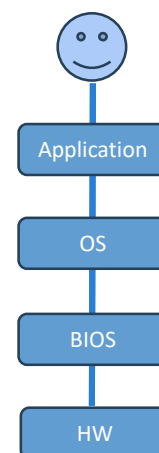


Modelos de Referência

Layering/Interfacing

- Layering is an important concept in computing.
- When a user runs an application,
 - The user interfaces with the application.
 - The application interfaces with the operating system.
 - The operating system interfaces with the BIOS.
 - The BIOS interfaces with the input/output hardware.



Layering advantages

- The user only needs to know something about the application and does not need to know any details about the operating system, the BIOS or the hardware.
- Similarly, the application needs information about the user and the operating system but not the BIOS or hardware.
- For layering to be successful, there must be a “standard” interface into a layer.
 - The operating system is said to provide an Application Programming Interface (API).

Layered industries

- These layers can be seen in the industry, there are
 - Companies that specialize in processors
 - Intel, AMD
 - Companies that specialized in components / boards
 - Asus, Seagate, Maxtor, Western Digital
 - Companies that specialize in BIOS
 - AMI, Phoenix
 - Companies that specialize in operating systems
 - Microsoft, Sun, B
 - Companies that specialize in applications
 - Oracle, Microsoft, AOL

Communications Layering

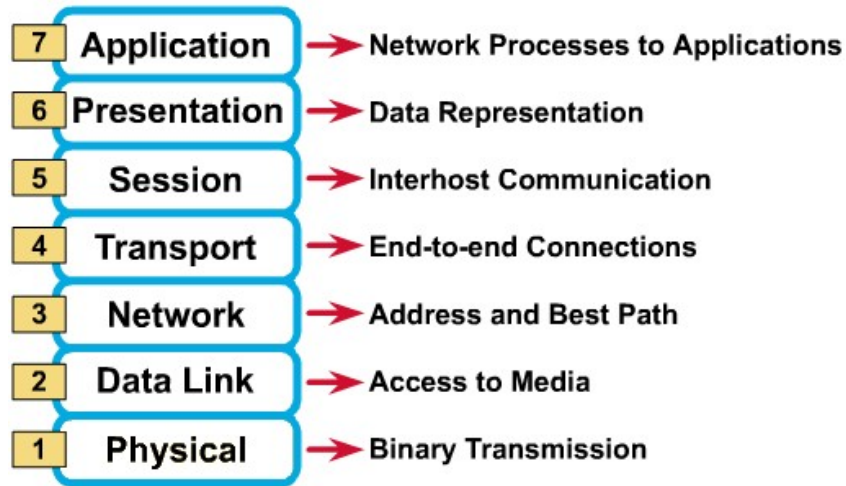
- The communication between two (or more) users on a network similarly is broken down into layers.
- A possible model for this layering is the:
Open Systems Interconnection model (OSI model)
- Laid out by :
 - by ISO (International Organization for Standardization).
 - as standard ISO/IEC 7498-1.
 - Adopted by CCITT/ITU



What is OSI?

- OSI serves as a "reference model" for how information is transmitted between any two points in a network.
- It guides manufacturers (a.k.a. vendors) so they can make products that will work with other products.
- The OSI model is comprised of seven layers that are involved in communicating between two nodes of a network.

OSI Model



Mnemonic

Please Do Not Throw Sausage Pizza Away

- **Please** → Physical
- **Do** → Data Link
- **Not** → Network
- **Throw** → Transport
- **Sausage** → Session
- **Pizza** → Presentation
- **Away** → Application



It's not the law

- The OSI model was never adopted as a rigorous set of specifications.
- The lower layers (physical, data link, network and transport) are for the most part adhered to.
 - Recall however that we have already mentioned a “brouter” which serves as both bridge (data link layer) and router (network layer).

Classroom not boardroom

- Many vendors agreed at one time or another to adhere to OSI, however, the system proved to be too loosely defined, overly broad and the companies had much invested in their proprietary standards.
- Some products like X.400 (an e-mail system used mainly in Europe and Canada, an alternative to SMTP) adhere to OSI.
- But mainly OSI has become a teaching tool.

OSI's Purpose

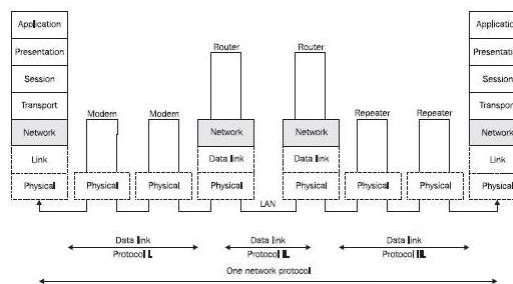
- Even if manufacturers do not restrict themselves to a particular OSI layer, they usually think of and describe their products in relationship to the OSI model.
- It provides a language and framework for discussion of networks, so it persists and continues to be taught.

The main idea

- The communication between computers can be broken down into layers.
 - Each layer is characterized by its functions and how it interfaces with the adjacent layers.
- Within the source's none and starting at the user level, the communication is passed down through the layers to the lowest layer where it is sent through some transmission medium.

The main idea

- The communication travels at the lowest layer (physical), occasionally rising up to the second layer (bridge) or third layer (router) until it reaches the destination.
- The communication now passes up through the layers.



The main idea (Cont.)

- The layers are implemented by a combination of the operating system, applications (for instance a browser), network protocols such as TCP/IP, and the software and hardware that actually place a physical signal onto a transmission line (NIC, modem, etc)

Layer 1: Physical

- A stream of 1's and 0's are put into a form convenient for transmission.
 - Waves (with little regard for their information content) are sent and received.
- This level is the most hardware oriented. It includes specifications about
 - NIC card speeds
 - Types and lengths of cable
 - Voltage characteristics (range, level or edge)
 - Etc.

Physical medium

- Wired
 - Copper
 - Cheaper cables and interfaces (+), suffers EMI (-)
 - Optical fibers
 - Immune to EMI, favors safety, wide bandwidth, low attenuation (+), expensive cables and interfaces (-)
- Wireless
 - Radio frequency
 - Mobility, flexibility (+), very susceptible to EMI (-), multi-path fading (-), attenuation (-), open medium (+/-)
 - Light
 - Mobility, flexibility (+), line-of-sight (-), open medium(+/-)

Layer 2: The data-link layer

- Synchronizes frames
 - Detects start of frame
- Converts signals into words of bits.
- Implements control flow

Layer 3: The network layer

- Implements routing of frames (packets) through the network.
- Defines the best path to send packets from the source to the destination.
- Defines logical addressing so that any endpoint can be identified.
- Handles congestion in the network.
- When required fragments a packet into smaller units to accommodate different media.

Layer 4: The transport layer

- Manages end-to-end communications.
- Service addressing.
- Implements a higher-order error-checking.
- Implements reliability
 - Acknowledgments.
 - Retransmission.

Layer 5: The session layer

- Establishes, manages and terminates sessions between applications /services.
- It is an agreement between a source and destination to communicate.
- Ensures that data units are delivered in sequence.

Layer 6: The presentation layer

- Provides independence from differences in data representation (e.g., encryption)
- Translates from application to network format, and vice versa.

Layer 7: The application layer

- Supports application and end-user processes.
- Provides services for transfers of application specific data packets:
 - Files
 - Emails
 - Commands
 - Pages
- Handles user authentication and privacy.

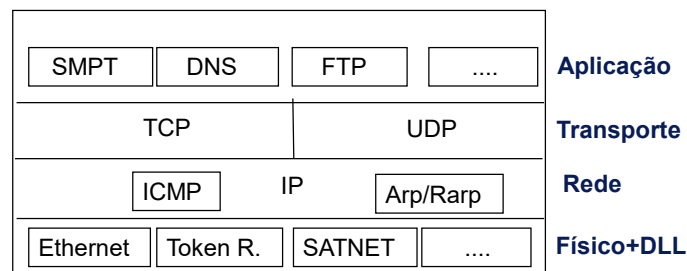
Modelo OSI

- OSI: Open Systems Interconnection
- Modelo de referência para comunicação de dados.
- Norma ISO: International Standards Organization
- Adotada pelo CCITT/ITU

TCP/IP model

- Another model for understanding communications networks is the TCP/IP model.
- It also is broken into layers
 - It has no equivalent of the OSI model's physical layer.
 - The next three layers of the OSI model and the TCP/IP model are roughly equivalent.
 - The top three layers of the OSI model are combined into one layer in the TCP/IP model.

Modelo TCP/IP



38

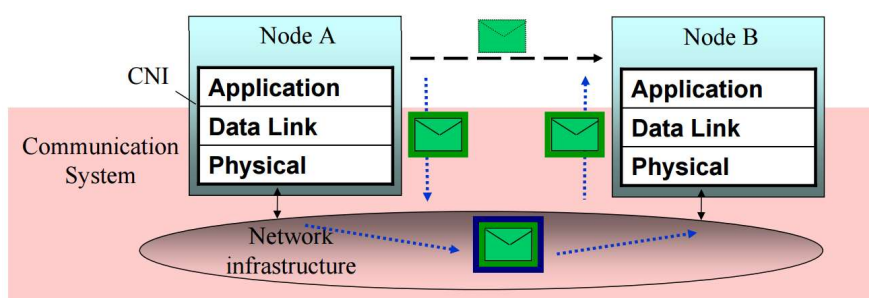
Real-time Networks

- The end-to-end communication delay must be bounded.
 - All services at all layers must be time-bounded.
 - Requires appropriate time-bounded protocols.
- The 7 layers impose a considerable computation and communication overhead...
 - The time to execute the protocol stack becomes dominant over communication time.
- Many real-time networks:
 - are dedicated to a well defined application
→ no need for presentation
 - use single broadcast domain
→ no need for routing
 - use short messages
→ no need to fragment/reassemble.

39

OSI collapsed model

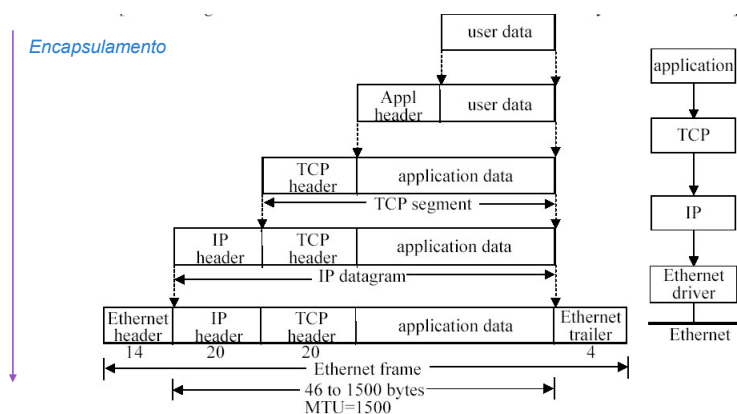
- Application services access the Data Link directly.
- Other layers maybe present but not fully stacked.
- In process control and factory automation these networks are called Fieldbuses.



[Luís Almeida,lda@det.ua.pt]

40

Encapsulamento



41

41

Modelo TCP/IP - Desmultiplexagem

Cabeçalho Ethernet (*type*)

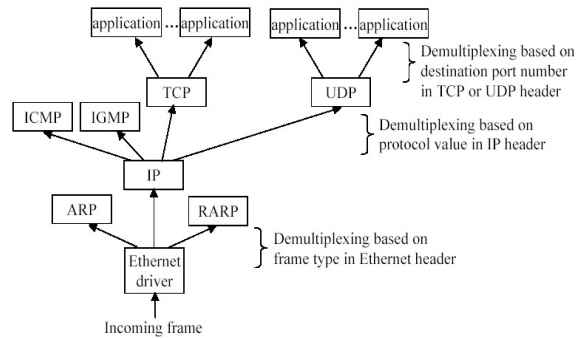
- IP 0x800
- ARP 0x0806
- RARP 0x8035

Cabeçalho IP (*protocol*)

- ICMP 1
- IGMP 2
- TCP 6
- UDP 17

Cabeçalho TCP/IP (*port*)

- FTP 21
- POP3 110



Physical Layer sublayers

- PCS: **Physical Coding Sublayer**
 - Block encoding
- PMA: **Physical Medium Attachment** sublayer
 - framing, octet synchronization/detection, and scrambling/descrambling
- PMD: **Physical Medium Dependent** sublayer
 - Adaption to the physical medium